

180. Consecutive numbers

PLATFORM

Platform: LeetCode

DIFFICULTY

MEDIUM

DATE

Solved: June 10, 2026

Problem Summary

Find all numbers that appear at least three times consecutively.

Return the result table in **any order**.

The result format is in the following example.

Key Insights

- Consecutive means adjacent rows based on id.
- Need to compare three neighboring rows.
- Same table can be joined multiple times to access neighboring records.
- If three consecutive rows have the same num, that number qualifies.

Observations

- We need row i , $i+1$, and $i+2$.
- Their num values must be identical.
- Multiple occurrences of the same qualifying number should appear only once in the output.

Algorithm

- Join Logs with itself as l2 where $l2.id = l1.id + 1$.
- Join another copy as l3 where $l3.id = l2.id + 1$.

$l1.num = l2.num$

$l2.num = l3.num$

- Filter rows where:
- Return distinct numbers.

Approach

- Treat the table as three copies: l1, l2, l3.
- Join l2 with the next row of l1.
- Join l3 with the next row of l2.
- Check that all three num values are equal.
- Use DISTINCT to remove duplicates.

Time Complexity & Space Complexity

TC : $O(n)$

SC : $O(1)$

SQL Concepts Used

Self join
Multiple Joins
DISTINCT
Filtering with WHERE

Database

Accepted

MySQL

Code

180. Consecutive Numbers

Medium

SQL Schema

Pandas Schema

Table: Logs

Column Name	Type
id	int
num	varchar

In SQL, id is the primary key for this table. id is an autoincrement column starting from 1.

Find all numbers that appear at least three times consecutively.

Return the result table in **any order**.

The result format is in the following example.

Example 1:

Input:

Logs table:

1	1
2	1
3	1
4	2
5	1
6	2
7	2

Output:

2.8K27470 Online

Accepted

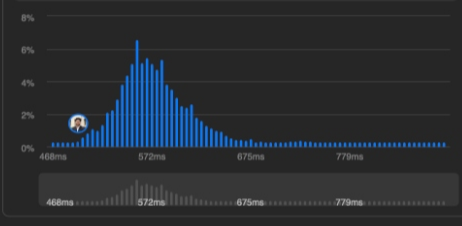
23 / 23 testcases passed

Sahil Khan submitted at Jun 10, 2026 13:29

Solution

Runtime

498 msBeats 99.50%



Code

MySQL

```
1 # Write your MySQL query statement below
2 SELECT DISTINCT l1.num AS ConsecutiveNums
3 FROM Logs l1
4 JOIN Logs l2
5 ON l2.id=l1.id+1
6 JOIN Logs l3
7 ON l3.id=l2.id+1
8 WHERE l1.num=l2.num AND l2.num=l3.num
```

Write your notes here

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Testcase

Test Result

You must run your code first